# SPLITNESS OF SIMPLICIAL OBJECTS IN SR(C) VIA DEGENERACY MAPS

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2766-2780 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 2782-2794, the complete author proof (11 body lines).

*Editorial note.* Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 22 of queue_hypercovering.json (pb=691, stmt 2766-2780, proof 2782-2794).

*Editorial note.* stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

*Editorial note.* Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-split' also exists in stacks-derived — a different proposition in a different chapter file.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 017P: sources/stacks-hypercovering/pinned-original-source.tex, source lines 2794–2794. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 017R: sources/stacks-hypercovering/pinned-original-source.tex, source lines 2794–2794. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $\mathcal{C}$ be a site. Let $K$ be an $r$-truncated simplicial object of $SR(\mathcal{C})$. The following are equivalent*

(1) *$K$ is split (Simplicial, Definition 017P),*
(2) *$f_{\varphi,i} : U_{n,i} \to U_{m,\alpha(\varphi)(i)}$ is an isomorphism for $r \geq n \geq 0$, $\varphi : [m] \to [n]$ surjective, $i \in I_n$, and*
(3) *$f_{\sigma_j^n,i} : U_{n,i} \to U_{n+1,\alpha(\sigma_j^n)(i)}$ is an isomorphism for $0 \leq j \leq n < r$, $i \in I_n$.*

*Date*: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*The same holds for simplicial objects if in (2) and (3) we set $r = \infty$.*

*Proof.* The splitting of a simplicial set is unique and is given by the nondegenerate indices $N(I_n)$ in each degree $n$, see Simplicial, Lemma 017R. The coproduct of two objects $\{U_i\}_{i\in I}$ and $\{U_j\}_{j\in J}$ of $\mathrm{SR}(\mathcal{C})$ is given by $\{U_l\}_{l\in I \amalg J}$ with obvious notation. Hence a splitting of $K$ must be given by $N(K_n) = \{U_i\}_{i\in N(I_n)}$. The equivalence of (1) and (2) now follows by unwinding the definitions. The equivalence of (2) and (3) follows from the fact that any surjection $\varphi : [m] \to [n]$ is a composition of morphisms $\sigma_j^k$ with $k = n, n+1, \ldots, m-1$. $\square$